

概率论总结. U2-3

随机变量:

伯努利, $X \in \{0, 1\}$

离散型

$$\begin{cases} \text{二项分布 } X \sim B(n, p), P(X=k) = C_n^k p^k (1-p)^{n-k} \\ E(X) = np, \text{Var} = np(1-p) \\ \text{泊松分布 } X \sim P(\lambda), P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!} \\ E(X) = \lambda, \text{Var} = \lambda \end{cases}$$

连续型

$$\begin{cases} \text{均匀分布 } X \sim U(a, b), \text{Var} = \frac{(b-a)^2}{12} \\ \text{指数分布 } X \sim E(\lambda), f(x) = \lambda e^{-\lambda x} (x \geq 0), E(X) = \frac{1}{\lambda} \\ F(x) = 1 - e^{-\lambda x}, \text{Var} = \frac{1}{\lambda^2} \end{cases}$$

正态分布 $X \sim N(\mu, \sigma^2)$ $f(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

标准正态: $f = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

- 一般正态 $\rightarrow \frac{X-\mu}{\sigma} \sim N(0, 1)$ 可查表.

$E = \mu, \text{Var} = \sigma^2$

$X \sim N(0, 1)$
 $\Rightarrow \sum_{i=1}^n X_i \sim N(0, n)$
 $X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$
 $\Rightarrow aX + bY \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2 + 2ab\rho \cdot \sigma_1\sigma_2)$

U4.

$= \int_{-\infty}^{\infty} x f_X(x) dx$

数学期望 $E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy$

$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy$
 $g(x, y)$

方差: $\text{Var}(X) / D(X) = E(X^2) - [E(X)]^2$

性质: $\text{Var}(\sum X_i) = \sum \text{Var}(X_i) + 2 \sum \text{Cov}(X_i, X_j)$

协方差: $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

性质: $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$

$\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$

相关系数 $\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$

$|\rho_{XY}| \leq 1$, XY 线性关系时取 1.

不相关 $\begin{cases} \text{Cov}(X, Y) = 0, \rho_{XY} = 0 \\ E(XY) = E(X)E(Y) \\ \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) \end{cases}$ (独立)

$\text{Cov} = 0$: 不相关. > 0 : 正相关.

$f(x)$: 概率密度

$P(X < x) = F(x) = \int_{-\infty}^x f(t) dt$ 分布函数

联合分布律 $P(X=i, Y=j)$

条件分布律 $P(X=i | Y=j) = \frac{P(X=i, Y=j)}{P(Y=j)}$

边缘分布律 $P(X=i)$

联合分布函数 $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) du dv$

边缘 $F(x) \Rightarrow f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$

条件 $f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = f_{Y|X}(y|x)$

独立 $\begin{cases} f(x, y) = f_X(x) \cdot f_Y(y) \\ F(x, y) = F_X(x) \cdot F_Y(y) \end{cases}$

多元: $z = X + Y$

$F_z(z) = \iint_{x+y \leq z} f(x, y) dx dy = \int_{-\infty}^z dv \int_{-\infty}^{v-u} f(u, v-u) du$

$f_z(z) = \int_{-\infty}^{\infty} f(x, z-x) dx = \text{独立} \rightarrow \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$

$z = g(X, Y): F_z(z) = \iint_{g(x, y) \leq z} f(x, y) dx dy$

$M = \text{Max}(X, Y): F_M(z) = F_X(z) F_Y(z)$

$\min: F_{\min}(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$

求 $E: \int \dots$

$\text{求 } E: \int \dots$

U5

大数定律: 趋近于概率, (宏观) 期望.

切比雪夫: 数学期望 μ , 方差 σ^2 .

$P\{|X - \mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}$

中心极限定理: 期望 μ , 方差 σ^2 的 $\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ 依概率收敛到期望.

$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma/\sqrt{n}} \xrightarrow{\text{分布}} N(0, 1)$

$\lim_{n \rightarrow \infty} P\left(\frac{\sum_{i=1}^n X_i - E(\sum_{i=1}^n X_i)}{\sqrt{\text{Var}(\sum_{i=1}^n X_i)}} \leq \frac{\sum_{i=1}^n X_i - n\mu}{\sigma/\sqrt{n}}\right) = \Phi(x)$

$P(X < a) = \Phi\left(\frac{a - \mu}{\sigma}\right) = \Phi\left(\frac{a - \mu}{\sigma}\right)$

关于 $\Phi: \Phi(-a) + \Phi(a) = 1$

$X_i \sim N(\mu, \sigma^2) \Rightarrow \sum_{i=1}^k X_i \sim N(k\mu, k\sigma^2)$
 $\uparrow$
 $\text{不是 } k^2$

16

总体 X $E(X)$ $Var(X)$ ① 若 $X \sim N(\mu, \sigma^2)$, 则 $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
 样本 $\sum X_i$ $E(\bar{X}) = E(X)$, $Var(\bar{X}) = \frac{Var(X)}{n}$ ② $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
 $E(S^2) = Var(X)$, $Var(S^2) = \frac{2\sigma^4}{n-1}$ ③ $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$, $\frac{S}{\sigma} \sim N(0, 1)$

χ^2 分布: $X_i \sim N(0, 1)$, $X = \sum X_i^2$, $X \sim \chi^2(n)$

$$\Rightarrow E(X) = n, Var(X) = 2n$$

④ $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$ ⑤ $\bar{X}, S^2$ 相互独立.

$$Y_1 + Y_2 \sim \chi^2(n_1 + n_2)$$

若 $Y \sim \chi^2(n)$, 查表得 $\chi^2_{\alpha}(n) = p \Rightarrow P(Y > p)$

$$P(\chi^2(n) > \chi^2_{\alpha}(n)) = \alpha \Rightarrow \text{上侧分位数} = p$$

t 分布: $t = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$, 其中 $X \sim N(0, 1)$, $Y \sim \chi^2(n) \Rightarrow t \sim t(n)$



F 分布: $U \sim \chi^2(n_1)$, $V \sim \chi^2(n_2) \Rightarrow F = \frac{U/n_1}{V/n_2} \sim F(n_1, n_2)$

$$\Rightarrow \frac{1}{F} \sim F(n_2, n_1)$$

若 $X \sim t(n)$, 则 $X^2 \sim F(1, n)$

若样本 $X_1 - X_{n1}$ 与 $Y_1 - Y_{n2}$ 分别来自 $N(\mu_1, \sigma_1^2)$ 和 $N(\mu_2, \sigma_2^2)$ 且相互独立, 则

$$\textcircled{1} F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$$

$$\textcircled{2} \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0, 1)$$

$$\textcircled{3} \sigma_1^2 = \sigma_2^2 = \sigma^2 \text{ 时, } \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

$$S_w^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$$

总体 $E(X) = \mu$ σ^2 θ
 样本 $\bar{X}$ S^2 $\hat{\theta}$

点估计

点估计: 无偏估计: ① $\mu_1 = E(X) = \dots \theta$ ② $\theta = \dots \mu_1 \Rightarrow$ ③ $\hat{\theta} = \dots \bar{X}$

极大似然: ① $L(\theta) = \prod_{i=1}^n f(X_i, \theta)$ ② $l(\theta) = \ln[L(\theta)]$ ③ $\frac{dl(\theta)}{d\theta} = 0$ ④ $\begin{cases} \text{若 } > 0, \min\{-\chi_n\} \\ < 0, \max\{-\chi_n\} \end{cases}$

评价准则: ① 无偏性: $E(\hat{\theta}) = \theta$, 关于自变量求出表达式求。

② 有效性: Var 小的无偏估计更有效。

③ 均方误差越小越好。 $Mse(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$ 若是无偏: $Mse = Var(\hat{\theta})$

④ 相合性: $\forall \varepsilon > 0$, 有 $\lim_{n \rightarrow \infty} P\{|\hat{\theta}_n - \theta| < \varepsilon\} = 1$ 成立。(大数定律/切比雪夫不等式)

满足正则条件时, 一定是相合的

$n \rightarrow \infty$ 时, 估计量方差 $\rightarrow 0$

$$\hat{\theta}_n \xrightarrow{P} \theta$$

求 $Var(\hat{\theta})$

若是 $\max$: 令 $Y = \max(\dots)$, $F_Y(y) = \int_{-\infty}^y f_Y(y) dy \Rightarrow (\dots)^n$

对 y 求导

$$f_Y(y) \rightarrow E = \int_{-\infty}^{\infty} y f_Y(y) dy$$

$$Var(\hat{\theta}) = [E(\hat{\theta}) - \theta]^2$$

区间估计



$\hat{\theta}_1$: 参数 θ 的置信度为 $1-\alpha$ 的区间估计

$\pm \hat{\theta}_2$

$\Rightarrow$ 参数 θ 的置信水平为 $1-\alpha$ 的置信区间

奈曼原则: 在一定置信度 $1-\alpha$ 下取最高精度的区间

选准变量: 单个正态总体 $X \sim N(\mu, \sigma^2)$

① σ^2 已知: $G = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
 ② σ^2 未知: $G = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$
 ③ μ 未知: $G = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$
 ④ μ 未知: $G = \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi^2(n)$

两个正态总体: 求 $\frac{\sigma_1^2}{\sigma_2^2}$ 置信区间: $G = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$

U8 假设检验.

第一类错误: 去真
= : 存伪

P 相互制约

$\bar{x} \geq c$ 时拒绝原假设 H_0
<

$$P_1 = \alpha(c) = 2 - 2\Phi\left(\frac{c}{\sigma/\sqrt{n}}\right)$$

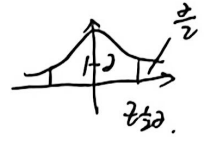
$$\Rightarrow c = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

奈曼-皮尔逊原则: 先控制犯第一类错误的概率 α , 再使第二类错误的概率 β 尽可能小.

α : 显著性水平. 显著性假设检验.

假设检验 " $H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$ " 中

$$H_0 \text{ 的拒绝域为 } W = \{|\bar{x} - \mu_0| \geq c = z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}\} \text{ 或 } W = \left\{ \left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| \geq z_{\frac{\alpha}{2}} \right\}$$



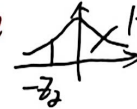
双边检验: $H_0: \theta = \theta_0, H_1: \theta \neq \theta_0$.

左边: $H_0: \theta \geq \theta_0$.

右边: $H_0: \theta \leq \theta_0$.

拒绝域 $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \leq -z_{\alpha/2}$

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \geq z_{\alpha/2} \text{ 右边}$$



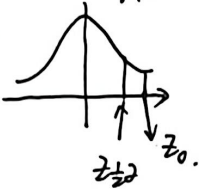
$P_- = P(|z| < |z_0|)$ 右边面积.

$P_- = P(|z| > |z_0|)$ 右边面积.

拒绝域和 H_1 是相符的.

P-值: $|z_0| = \left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right|, P_- = P(|z| > |z_0|)$

$\Rightarrow$ 更极端,
更向 H_1 .



$P_- < \alpha$ 显著性水平: 样本数据落在拒绝域内, 拒绝原假设

$\Rightarrow$ 在水平 α 下统计显著.

左边: $P_- = P(z < z_0) = \Phi(z_0)$

右边: $P_- = P(z > z_0) = 1 - \Phi(z_0)$

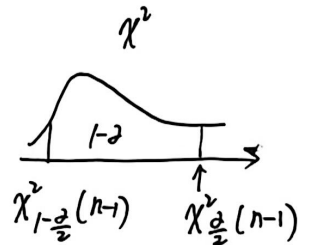
解题步骤: ① $H_0 =$ $H_1 =$

② 检验统计量: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; \chi^2 = \frac{(n-1)s^2}{\sigma^2}$

③ H_0 的拒绝域为: eg $T > t_{\alpha}(n-1)$

④ 代入, 是否在拒绝域内

⑤ 结论.



$f(x, y)$: 要分类讨论 ($=0$ 时); 画图 (xy 范围)